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DATA RESOURCE

SIMPLER: the Swedish infrastructure for medical population-based life-course and environmental research

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Abstract

The Swedish Infrastructure for Medical Population-based Life-course and Environmental Research (SIMPLER) is a national research infrastructure that provides longitudinal data and biological samples to investigate how dietary, lifestyle, environmental, and genetic factors influence health and disease across the life course. SIMPLER integrates two population-based prospective cohorts: the Swedish Mammography Cohort and the Cohort of Swedish Men. Baseline data were collected between 1987 and 1990 among 61,433 women and in 1997 among 45,906 men residing in central Sweden. Participants have completed repeated questionnaires on diet, lifestyle, and health approximately every ten years, complemented by linkage to national health registers providing information on diagnoses, mortality, prescribed medications, and socioeconomic status. Two comprehensive subcohorts nested within the cohorts (n = 8046 women, n = 4773 men) provide data from detailed health examinations, questionnaires, and repeated collection of blood, saliva, urine, stool, and adipose tissue biopsy samples. Multi-omics data are available for approximately 12,800 participants, and genomic data for about 51,000 individuals. By integrating long-term questionnaire data, register linkages, and extensive biological sampling, SIMPLER supports research on disease prevention, biomarker discovery, and precision health strategies.

Keywords Cohort study · Diet and lifestyle · Multi-omics · Register data · Swedish Mammography Cohort · Cohort of Swedish Men

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Introduction

Chronic late-onset diseases, such as cardiovascular disease, cancer, fragility fractures, osteoarthritis, type two diabetes, and cognitive health, are partly preventable burdens. Longitudinal study designs are crucial for elucidating factors involved in the prevention, prediction, prognosis, and treatment of chronic late-onset diseases. Prospective cohort studies that leverage time-updated exposure data, register-based outcomes, and biological sample collection are essential for disentangling the complex factors underlying disease development. The Swedish Infrastructure for Medical Population-based Life-course and Environmental Research (SIMPLER) is a national research infrastructure that supports such epidemiological investigations. SIMPLER presently comprises two cohorts: the Swedish Mammography Cohort (SMC) and the Cohort of Swedish Men (COSM). These cohorts are based on longitudinal assessments of diet, other lifestyle factors, and virtually all disease outcomes and causes of death. Repeated measurements of these factors over time may reduce information bias and provide unique opportunities to investigate temporal changes and detailed dose–response relationships, and to assess critical exposure periods.

Initial data collection for the cohorts began more than 30 years ago. Baseline data for women in SMC were collected between 1987 and 1990, and data collection for men in COSM has been ongoing since 1997. Over time, the scope of investigations has expanded to include health examinations and repeated collections of diverse biological samples (blood, saliva, urine, stool, and adipose tissue biopsies) in two substudies nested within the main cohorts. In addition, linkage to 20 national registers enables the study of a wide range of diseases.

Current research focuses on determinants of health and disease, include genetic, environmental, toxicological, diet and lifestyle, behavioural and societal factors, as well as gene-environment and gene-lifestyle interactions. The infrastructure further allows investigation of individual disease trajectories, the effects of co-morbidity, multimorbidity, and polypharmacy, and the identification of biomarkers for early diagnosis and potential drug targets. Moreover, data generated by omics technologies provide insights into both exogenous exposures, such as diet, and endogenous biological responses to them. An integrated multi-omics approach based on multiple sample types offers a holistic view of molecular complexity between individuals and can be used to inform tailored strategies for the prevention and treatment of chronic late-onset diseases [1].

A 2013 publication described the initial study design, methods, and characteristics of SMC (n = 61,433) and COSM (n = 45,906), including their respective subcohorts (n = 8046 women, n = 4773 men) [2]. The present paper has two aims: (1) to provide an overview of the substantially expanded cohort follow-ups and their integration into SIMPLER, and (2) to describe the newly available data within the comprehensive subcohorts, together with their associated methods, data collection procedures, and analytical approaches, including multi-omics. In addition, the paper briefly illustrates how SIMPLER data have been used in various epidemiological studies.

The Swedish Infrastructure for Medical Population-based Life-course and Environmental Research (SIMPLER)

The SIMPLER database and biobank

SIMPLER is a national research infrastructure comprising a database with individual-level participant data from SMC and COSM, and a biobank containing samples from the substudies of these cohorts, referred to as the SMC Comprehensive subcohorts (SMCC Uppsala and SMCC Västmanland) and the Västmanland-based COSM Comprehensive subcohort (COSMC). The main cohorts are collectively referred to as the SIMPLER cohorts. The database includes questionnaire data, health examination results (e.g., physical performance and memory tests), body composition measurements, biochemical and omics data, and information obtained through linkage to national registers. Since 2018, SIMPLER has been sustained and enriched through funding from the Swedish Research Council and its four institutional members: Uppsala University, Karolinska Institutet, Örebro University, and Chalmers University of Technology. The infrastructure is hosted by Uppsala University and governed by a Consortium Collaboration Board consisting of expert representatives from these four institutions and other Swedish universities. Additional support is provided by the County Councils of Uppsala, Västmanland, and Örebro. Detailed information on available data and infrastructure resources is provided on the SIMPLER website (www.simpler4health.se). SIMPLER supports both hypothesis-driven and data-driven epidemiological research by offering extensive, well-characterized data resources, i.e., a smorgasbord of data. Figure 1 provides an overview of data, biobank samples, and their roles in various lines of research.

Data harmonization and international consortia

Researchers in Sweden and internationally have used data from SMC and COSM in major diet- and health-related research projects since 1992. Questionnaire-based data on physical activity and diet have, for example, been

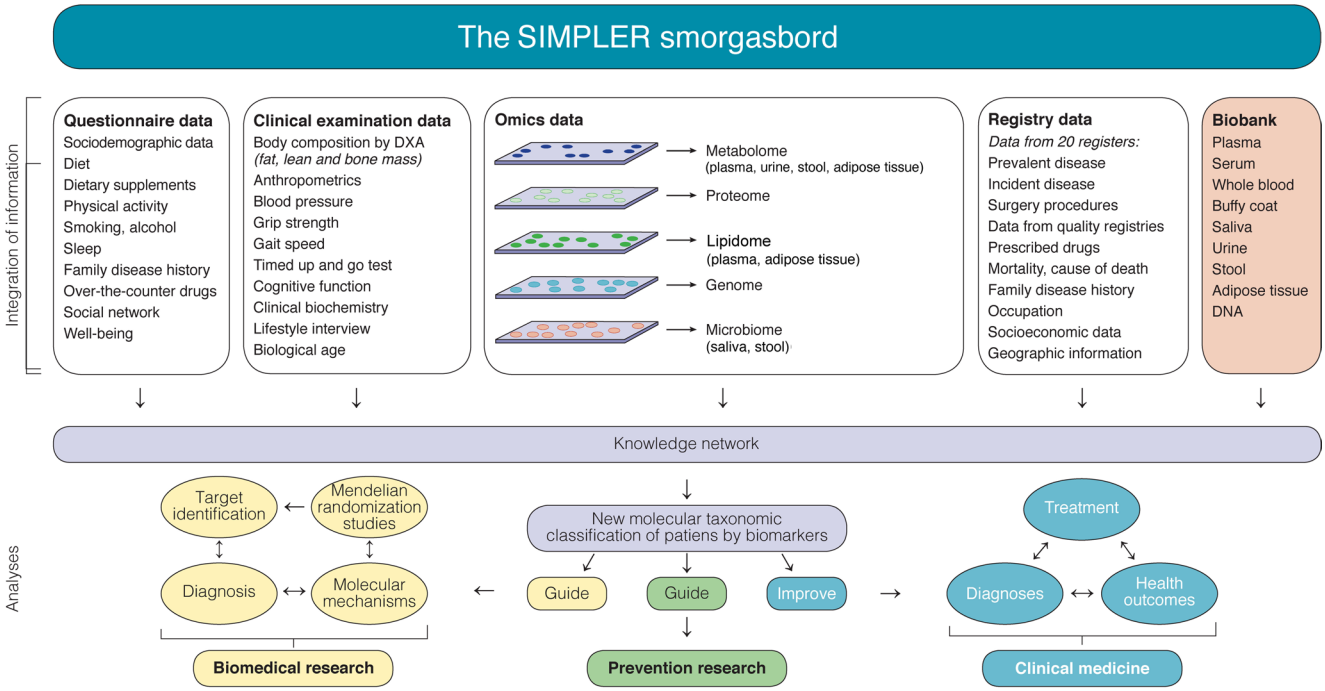


Fig. 1 The Swedish Infrastructure for Medical Population-based Life-course and Environmental Research (SIMPLER) comprises a database with individual participant data from Swedish Mammography Cohort (SMC) and Cohort of Swedish Men (COSM) and a biobank with samples from the comprehensive subcohorts, SMCC and COSMC

harmonized with corresponding data from other leading research institutions, including Harvard University (USA), Oxford University, University of Cambridge, and Imperial College London (UK), as well as the National Cancer Institute (USA), thereby facilitating international collaboration. SIMPLER participates in several international consortia that contributed data on multiple cancer types, including breast, ovarian, prostate, gastric, colorectal, and renal cell cancer, as well as on other diseases such as diabetes and osteoarthritis, together with genetic data, lifestyle factors (physical activity, diet, etc.), and health-related measures. The cohorts are registered at ClinicalTrials.gov (Identifier NCT01127698 (SMC)) and NCT01127711 (COSM)).

Data access

SIMPLER provides a digital platform for national and international researchers to apply for data access. Detailed information on data access procedures is available on the SIMPLER website (www.simpler4health.se). Data for approved projects are delivered to a secure, project-specific workspace hosted on the sensitive data server of the National Academic Infrastructure for Supercomputing in Sweden, NAISS (https://www.naiss.se/). The NAISS-SENS server is operated by Uppsala Multidisciplinary Center for Advanced Computational Science, UPPMAX, at Uppsala University, Uppsala, Sweden. Access to the secure workspace requires two-factor authentication, and analyses are conducted in an environment designed for sensitive personal data that is in compliance with national personal data security regulations and the European General Data Protection Regulation (GDPR).

Study design

Baseline questionnaire data were collected between 1987 and 1990 among women in the Swedish Mammography Cohort (SMC) and in 1997 among men in the Cohort of Swedish Men (COSM). Thereafter, participants completed additional lifestyle and health questionnaires approximately every 10 years during follow-up. Data collection is still ongoing in both the main cohorts and subcohorts (described below), with the next follow-up in the main cohorts scheduled for completion in 2028. Participation flow in the cohorts and subcohorts is illustrated in Fig. 2a and b.

Questionnaire surveys in the main study cohorts

At cohort entry, participants in the main study cohorts, SMC and COSM, resided in three adjacent counties in central Sweden (Uppsala, Västmanland, and Örebro). Between 1987 and 1990, all women born between 1914 and 1948 (age range 38–76 years) living in Uppsala or Västmanland counties were invited to participate in a mammography

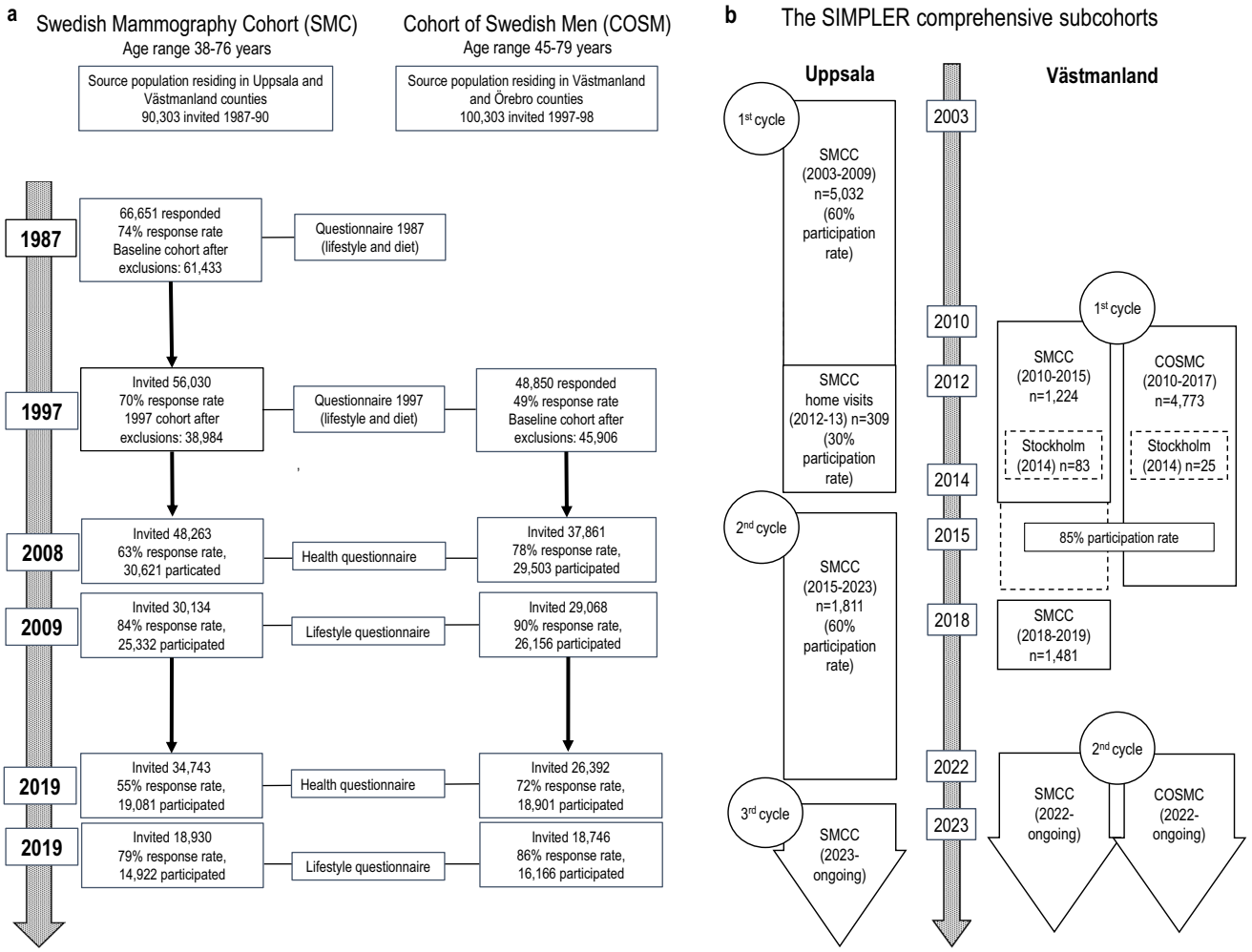


Fig. 2 Data collection in SIMPLER. **a** Flow chart of the data collection in the main study cohorts, Swedish Mammography Cohort (SMC) and Cohort of Swedish Men (COSM). Exclusions done in SMC 1987–90 were (1) incorrect or missing national registration number, (2) pre-baseline cancer diagnosis (except non-melanoma skin cancer), (3) implausible energy intake (±3 SD from the log-transformed mean intake), (4) incompleted questionnaires. For COSM 1997 exclusions were the same as for SMC 1987–90 except for extreme energy intakes, and the major reason for exclusion was pre-baseline cancer. The number of respondents and response rate (%) are visualized. **b** Flow chart of the data collection at the different examination cycles in the comprehensive subcohorts, SMCC and COSMC. The participation rate given reflects the total number of participants. Subcohort participants were mostly selected randomly from the main cohorts. SMCC, Swedish Mammography Cohort Comprehensive subcohort; COSMC, Cohort of Swedish Men Comprehensive subcohort

screening programme and to complete a questionnaire focusing on their diet, as described below. Seventy-four percent of eligible women agreed to participate and completed the questionnaire (n = 61,433 after exclusions), forming the basis of SMC.

In autumn 1997, a second, more extensive questionnaire was distributed to all SMC participants in the study area. Concurrently, all men born between 1918 and 1952 (age range 45–79 years) and residing in Västmanland or Örebro counties were invited to participate in a similar lifestyle questionnaire study. Forty-nine percent of eligible men agreed to participate and completed the questionnaire (n = 45,906), forming the basis of COSM. Participants in SMC and COSM completed additional rounds of questionnaires in 2008 and 2009, and twice in 2019, as described below.

Since 1987, around 3000 participants have migrated within Sweden but outside the study areas, which are shown in Supplementary Information 1, and around 500 have formally emigrated. Consequently, fewer than 0.5% of participants have been completely or partially lost to follow-up, since register-based data remained available for all individuals residing in Sweden. Notably, a proportion of participants who formally emigrated continue to appear in Swedish registers after the date of emigration, particularly with respect to healthcare utilization and prescription data.

Saliva sample collection and processing in the main study cohorts

All participants in COSM and a subset of SMC participants, due to financial constraints, were invited to provide saliva samples for DNA extraction using a home-based collection kit between 2005 and 2009. Samples were collected using Oragene OG-250 (DNA Genotek; Ottawa, Ontario, Canada, maximum volume 2 mL). Approximately 40,000 samples were returned by regular mail and stored at − 20 °C in the Biobank. Sex of sample providers was verified by genetic analyses.

Biological sample collections and examinations in the comprehensive subcohorts

SMCC and COSMC were initiated in 2003 and 2010, respectively, to collect health examination data, biological samples, and additional lifestyle and health information through questionnaires. SMCC comprises women residing in Uppsala or Västmanland counties, whereas COSMC includes men exclusively from Västmanland or Örebro counties. Accordingly, both men and women were examined at the Västmanland research clinic in the city of Västerås, while only women were examined at the research clinic in the city of Uppsala (Supplementary Information 1). Table 1 provides an overview of the tests and measurements conducted in SMCC and COSMC. Recruitment of the participants to SMCC and COSMC was mostly random, and the samples size was mainly limited by financial reasons. In Västmanland, older participants were targeted initially, and participants who additionally were participating in another population-based cohort study [3] were prioritized, as were cohort participants from the same family.

SMCC Uppsala

Between 2003 and 2013, SMC participants born between 1920 and 1940 and residing in Uppsala County (n = 8311) received a lifestyle questionnaire and an invitation to a health examination 1–3 months after questionnaire completion. Among participants younger than 85 years, health examinations were conducted at the Uppsala research clinic, with a participation rate of 60% (n = 5032). Participants aged 85 years or older underwent health examination in their homes during 2012–2013; of the 1030 women invited, 309 (30%) agreed to participate. This subcohort is referred to as SMCC Uppsala and comprises a total of 5341 women.

COSMC and SMCC Västmanland

COSMC and SMCC in Västmanland comprise 4773 men and 2705 women, respectively, for a total of 7,478 participants. Between 2010 and 2017, all male participants in COSM born between 1920 and 1952 residing in Västmanland County were invited to participate in COSMC using the same collection protocol as in SMCC in Uppsala.

Concurrently (2010–2015), SMC participants who were spouses or partners of COSMC participants living in Västmanland County were invited to join SMCC Västmanland. In total 1062 couples attended the health examination. Additional SMC women who were not partners of the COSMC participants were recruited to SMCC Västmanland during 2018–2019. Participants in SMCC Västmanland and COSMC participants aged 85 years or older, or with health-related limitations, underwent health examinations during home visits during the same period. Altogether, 85% of those invited in Västmanland County underwent the health examinations. A subgroup of SMC and COSM participants (n = 108) underwent the health examination at Karolinska Institutet in Stockholm in 2014 and are included in the cycle 1 totals for SMCC Västmanland and COSMC. These participants were originally residents of Västmanland County but had relocated to Stockholm by the time they were invited to participate in the Västmanland subcohorts.

Subcohort examination cycles

Data and biological samples have been collected repeatedly over time in the subcohorts through a process referred to as examination cycles (further described below). The first examination cycle was undertaken between 2003 and 2009 for SMCC Uppsala, and between 2010 and 2019 for COSMC and SMCC Västmanland. The second examination cycle for SMCC Uppsala was conducted from 2015 to 2023. All participants residing in Uppsala County who had completed the health examination and questionnaires during the first cycle were invited to participate in the second cycle; 60% agreed (n = 1811). The mean interval between the first and second examination cycles was 12 years. In 2022, a second examination cycle was initiated in COSMC and SMCC Västmanland and is still ongoing. A third examination cycle in SMCC Uppsala began in 2023 and is also ongoing.

Register data

SIMPLER holds multiple ethical approvals for linkage to several Swedish registers. Register data are linked to participants through their national personal registration number, ensuring no loss to follow-up as long as participants remain in Sweden. These data include information on

Table 1 Tests and measurements performed at the health examinations at research clinics and home visits in the subcohorts, Swedish Mammography Cohort Comprehensive subcohorts (SMCC) and Cohort of Swedish Men Comprehensive subcohort (COSMC)

Research clinic			City of UPPSALA					City of VÄSTERÅS							
		Total	SMCC ♀				Total	SMCC ♀				COSMC ♂			
Recruitment round			CYCLE 1		CYCLE 2	CYCLE 3			CYCLE 1		CYCLE 2		CYCLE 1		CYCLE 2
			RC	HOME				Total	RC	HOME		Total	RC	HOME	
Collection period			Sep 2003–Sep 2009	Oct 2012–Mar 2013	Feb 2015–Jun 2023	Dec 2023 →			Apr 2010–Jun 2019	Mar 2013–Apr 2015	Oct 2022 →		Apr 2010–Jan 2018	Mar 2013–Apr 2015	Oct 2022 →
Number of participants	12,819	5341	5032	309	1811	Ongoing	7478	2705	2672	33	Ongoing	4773	4720	53	Ongoing
Anthropometry[a]			√	√	√	√			√	√	√		√	√	√
DXA	Lunar prodigy		√		√	√					√				√
Bioimpedance	Tanita, BC-418								√						
Supine blood pressure			√	√	√	√			√	√	√		√	√	√
Supine heart rate				√	√	√			√	√	√		√	√	√
Cognitive tests[b]				√	√	√			√	√	√		√	√	√
Functional tests[c]				√	√	√			√	√	√		√	√	√
Biological samples															
Blood			√	√	√	√			√	√	√		√	√	√
Urine			√	√	√	√			√	√	√		√	√	√
Adipose tissue			√	√	√	√			√	√	√		√	√	√
Stool				√	√	√			√	√	√		√	√	√
Saliva	Collected in the main cohorts							Collected in the main cohorts							

RC research clinics; *HOME* home visits; *DXA* dual-energy X-ray absorptiometry

[a]Weight, height, waist circumference, hip circumference, upper arm circumference, calf circumference, and sagittal abdominal diameter

[b]Telephone-based interview [24] and since 2014 the Mini-Mental State Examination at the examination [25]

[c]Chair stand, grip strength, and gait speed

clinical diagnoses, causes of death, drug prescriptions, and socioeconomic status. The registers are maintained by the Swedish National Board of Health and Welfare, Statistics Sweden, and national quality registers. Detailed information on the specific registers available is presented on the SIMPLER website (https://www.simpler4health.se/).

Data and sample collection in the main cohorts

Questionnaires formed the foundation of data collection in the cohorts. Initially administered in paper format, responses were first entered manually into a database and later processed using automated data entry methods. Participants were not re-contacted to clarify any ambiguous responses. The questionnaires are briefly described below, and more detailed information is available on the SIMPLER website (https://www.simpler4health.se/). Since 1997, questionnaire data collection in both SMC and COSM has been repeated approximately every 10 years, and over time, the questionnaires have increased in scope and number of items.

Questionnaires 1987 and 1997

The baseline questionnaire administered to SMC participants in 1987–90 comprised 80 items and collected information on socio-demographic characteristics, anthropometric measures (height, weight, and waist and hip circumference), dietary intake (see section on Dietary Intake Assessment), alcohol consumption, and family history of breast cancer. The expanded questionnaire administered in 1997 included approximately 190 items and additionally captured information on dietary supplement use, medication, smoking, physical activity, sleep habits, medical history, and family history of breast cancer. The 1997 questionnaire was identical for men in COSM as for women in SMC, except for a small number of sex-specific items (e.g., prostate-related medication uses and gynaecological history, respectively).

Health questionnaires 2008 and 2019

In 2008 and 2019, a health questionnaire was distributed to all living study participants. The questionnaires were identical and comprised approximately 100 items covering anthropometric measures, medication use, sleep habits, bowel habits, self-rated health, well-being [4], stress, social network, pain, hearing, vision, balance, dental health, disease history, surgeries, and family history of disease.

Lifestyle questionnaire 2009 and 2019

In 2009, all participants who had completed the 2008 health questionnaire were invited to complete a lifestyle questionnaire. Similarly, all participants who had completed the 2019 health questionnaire were invited to complete a lifestyle questionnaire later that same year. The lifestyle questionnaire comprised approximately 230 items, covering dietary intake, alcohol consumption, dietary supplement use, medication use, physical activity, smoking, outdoor activities, and sun exposure habits.

Dietary intake assessment

Dietary intake was assessed using a food frequency questionnaire (FFQ) included in the 1987, 1997, 2009, and 2019 lifestyle questionnaires, comprising 67, 96, 149, and 148 food items (including alcohol and bread spreads), respectively. Participants reported their average consumption frequency for each food item over the preceding six months (1987 FFQ) or the past year (1997, 2009, and 2019 FFQs) to assess habitual dietary intake. Responses were recorded using eight predefined frequency categories ranging from "never/seldom" to "4 (FFQ 1987) or 3 or more times per day" (later FFQs). Frequently consumed foods (e.g., dairy products and bread) were also reported as daily servings using open-ended questions.

The increased number of food items in the later FFQs reflects efforts to collect more detailed information on food groups and the expanding diversity of foods available over time on the market. For example, for cheese, which from the beginning was one item in the 1987–90 FFQ (cheese unspecified), two items in 1997 FFQ (cheese (not low-fat) and low-fat cheese), and five items in 2009 and 2019 FFQ (hard cheese (not low-fat), low-fat hard cheese, cheese spread (not low-fat), low-fat cheese spread, and other cheese ("dessert ost")). The items can be co-analysed between the questionnaires, but due to the differing number of items between the questionnaires, we advise careful data management. Energy and nutrient intakes were estimated by multiplying reported consumption frequency of each food item by the nutrient values derived from sex- and age-specific portion sizes. Portion sizes were based on SIMPLER's internal validation studies, and later also portion size data from the Swedish national dietary survey Riksmaten adults 2010–11 [5]. Nutrient and energy values were obtained from food databases from the Swedish Food Agency: the 1992 database for the 1987 FFQ, the 1997 database for the 1997 FFQ, and the 2012 database for the 2009 and 2019 FFQs. The 2009 and 2019 questionnaires additionally assessed meal patterns (breakfast, lunch, dinner, and snacks) and portion sizes using six photographic images. The SIMPLER diet database contains information on total energy intake and intake of 51 nutrients. The estimated intake of each nutrient [6] is available both as an absolute value and as an energy-adjusted value calculated using the residual method [7].

Participants with implausible energy intakes (±3 standard deviations from the sex-specific mean value of the natural logarithm-transformed energy intake) were excluded from the energy-adjusted analyses.

Validation studies

Validation studies within SIMPLER have been undertaken to evaluate the accuracy of questionnaire-based data [6, 8–18] and are described in the Methods documentation under Questionnaire data collection at the SIMPLER webpage [19]. Estimated nutrient intakes derived from the FFQs have been evaluated for validity and reproducibility. FFQ-based estimates, compared to repeated 24-h recall interviews, yielded deattenuated correlation coefficients ranging from 0.44 to 0.81 (mean 0.65) for macronutrients and from 0.25 to 0.77 (mean 0.49) for micronutrients [6]. Reproducibility, assessed using a test–retest design, showed intra-class correlation coefficients ranging from 0.54 to 0.85 across all nutrients [6]. Biomarker-based validation studies further support the accuracy of dietary assessment. Using adipose tissue fatty acid composition as an indicator of long-term intake, FFQ-derived estimates of polyunsaturated fatty acids in women [17], long-chain n-3 fatty acids in women [15, 20] and in men [20], linoleic acid, alpha-linolenic acid and myristic fatty acid in men and women [20], as well as dairy fat [16, 18] showed reasonable validity, with correlation coefficients ranging from 0.30 to 0.65. Validation studies of milk intake demonstrated correlations of approximately 0.70 between FFQ estimates compared to four 7-day food records [21]. Consistent with these findings, a positive correlation of approximately 0.60 was observed in both sexes between reported milk intake and the adipose tissue content of pentadecanoic acid, a biomarker reflecting long-term intake of milk fat present in both milk and fermented milk products [18, 22]. In addition, repeated questionnaires capture documented population-level changes in dairy consumption patterns (non-fermented and fermented milk intakes) over time [23].

Exercise and physical activity are assessed using seven separate questions. The exercise question, with six predefined response categories ranging from never to more than 5 h per week, has demonstrated acceptable validity, with concordance correlation of 0.64 when compared to activity records and 0.38 when compared to accelerometer-based estimates, as well as good reproducibility, with an intraclass correlation of 0.69 based on a one-year apart test–retest assessment [12, 13].

Data collection in the comprehensive subcohorts

Questionnaires

On average, participants in the subcohorts completed a separate lifestyle questionnaire approximately one month prior to the health examination, except during the first cycle of COSMC and SMCC Västmanland, when both a lifestyle and a health questionnaire were administered together. All questions referred to the preceding month. Due to ongoing questionnaire development, several versions of the lifestyle questionnaire were used during the first examination cycle of SMCC Uppsala. The development process resulted in the standardized lifestyle questionnaire, subsequently used to follow up the main cohorts in 2009 and 2019. The development of the questionnaire is further explained in Supplementary Information 2. The final version has also been used in the first and second cycles of COSMC and SMCC Västmanland and in the second and third cycles of SMCC Uppsala. Harmonized variables and a detailed description of the questionnaire are available on the SIMPLER website (https://www.simpler4health.se/). From 2010 onwards, participants additionally completed a one-day food record. From 2013 onwards, participants were also asked to complete a toilet habit questionnaire, including the Bristol stool scale.

Health examination

Participants attended the research clinic in Uppsala or Västerås after overnight fasting to undergo a comprehensive health examination. They brought completed questionnaires and self-collected stool and urine samples, collected at home using standardized kits and instructions (see below). An experienced research nurse used standard equipment to measure blood glucose, HbA1c, heart rate, height, weight, hip and waist circumference, upper-arm and calf circumference, and sagittal abdominal diameter. Blood pressure was measured twice in the supine position using a digital blood pressure monitor to the nearest mm Hg after at least 10 min of rest. In Uppsala, body composition and bone mineral content were measured by dual-energy X-ray absorptiometry (DXA; Lunar Prodigy, Lunar Corp., Madison, WI, USA), yielding estimates of lean body mass, fat mass, bone mineral density, bone mineral content, and bone mineral area (hip region, spine, and total body). In addition, lateral vertebral assessment (LVA) has been used to assess vertebral fractures in the thoracic and lumbar spine. All DXA measurements were performed by the same research nurse, using the same equipment throughout the study in Uppsala. Precision errors derived from triple DXA scans on 15 participants, including repositioning, ranged from

0.8% to 1.5%, depending on the measurement type (lean mass, fat mass, or bone mineral density) and body region. The long-term coefficient of variation since 2003 was less than 1% for a spine phantom. In Västerås, body composition was measured using bioelectrical impedance analysis (Tanita, BC-418), providing estimates of body fat mass, fat-free mass, and body fat ratio in approximately 1000 women. From 2022 onward, body composition, including bone mineral density, has also been assessed in Västerås using DXA (Lunar Prodigy). The health examination further included functional tests (hand grip strength by use of Smedley's dynamometer (100 kg, Scandidact, Denmark), the Chair Stand Test, and 8-foot walk speed) and a structured health interview addressing physical activity and functional status, vision and hearing, dental health, sleep habits, weight loss, depressive symptoms, and quality of life. Between 2010 and 2014, cognitive function was additionally assessed using a telephone-based cognitive test [24]. Since 2014, the Mini-Mental State Examination has been incorporated into the examination visit [25]. More information is available at https://www.simpler4health.se/.

Sample collection during the health examination

Blood and adipose tissue biopsy samples were collected during the morning health examination at the research clinic following an overnight fast. All samples were protected from light during collection and processing and were stored at − 70 °C or lower within 45 min. EDTA plasma, lithium/heparin plasma, citrate plasma, serum, and buffy coat were obtained from whole blood under standardized centrifuging conditions and aliquoted into 500 µl cryotubes according to established protocols [26]. Stool samples were self-collected at home up to three days before the visit (on average, one day prior), stored at − 20 °C (normal freezer temperature in homes in Sweden), and then transported to the research clinic on the day of the examination. Further, on the day of the clinic health examination, participants collected a first-morning void urine sample at home.

All biological samples were longterm stored in ultra-low-temperature freezers from the time of collection at the clinic or when they arrived at the clinic until analysis. From 2003 to 2023, samples were stored at − 80 °C. In line with international initiatives to reduce the carbon footprint of biobanks, freezer temperatures were raised to − 70 °C in 2023 [27, 28]. Before 2003, there was no collection of biological samples in these cohorts.

Sample analyses

Biochemical analyses

Collected samples have been used for a wide range of ordinary biochemical analyses. Blood samples were analyzed for glucose, HbA1C (both directly measured at a benchtop analyser), insulin, lipid profile, C-reactive protein, biomarkers of renal- and liver function, iron and transferrin, calcium homeostasis markers, bone turnover, thyroid-stimulating hormone, albumin, and electrolytes. Urine samples were analyzed for essential elements, creatinine, isoprostanes, prostaglandins, and contaminants, and adipose tissue biopsy samples were analyzed for fatty acid composition [29]. Biochemical analyses were performed across multiple collection cycles in SMCC and COSMC subcohorts. Information on analytical coverage and detailed descriptions of laboratory methods are provided in the variable list available in Supplementary Information 3 and on the SIMPLER website (www.simpler4health.se). Most analyses were conducted at Department of Clinical Chemistry and Pharmacology, Uppsala University Hospital, in 2014 and 2020, and at Uppsala Biobank (Uppsala BioLab) in 2023.

Omics measured in blood, saliva, urine, stool, and adipose tissue

The omics component of the SIMPLER database comprises data from genomics, proteomics, metabolomics, lipidomics, and metagenomics analyses. With the exception of genomics, which was additionally performed in a subset of the main cohorts using saliva-derived DNA (n = 40,000), all omics analyses were conducted using samples from the SMCC and COSMC. A summary of the omics analyses, including the sample numbers for analysis is available in Table 2. In total, approximately 12,800 participants contributed biological samples for one or more omics analyses.

Genomics

DNA has been extracted from blood and saliva samples, yielding a total of approximately 53,000 samples, with around 2000 individuals overlapping between the two sample types. Genome-wide genotyping of single nucleotide polymorphisms (SNP) was performed using the Infinium Global Screening Array Multiple Disease v1.0 (GSA-MD, Illumina Inc, San Diego, CA, USA) for SMCC Uppsala and the newer and updated Infinium Global Screening Array v3.0 (GSAv3) for the remaining DNA analyses. Genetic analyses in SMCC and COSMC are based exclusively on whole blood [30]. In addition, a subset of saliva samples was genotyped using the Infinium Oncoarray − 500 K

Table 2 Summary of number of individuals with omics-analyses completed, ongoing or planned for, in the Swedish Mammography Cohort Comprehensive Subcohort (SMCC) and Cohort of Swedish Men Comprehensive Subcohort (COSMC)

Status	Type of omics	Sample type	Analysis center	Technology, other information	Number of individuals
Completed	Genomics, main cohorts	Blood, saliva	SciLifeLab/Eurofins	Illumina Global Screening Array	53,000[a]
	Untargeted metabolomics, LC–MS	Plasma	Chalmers University of Technology	5000 metabolic features	12,800 (cycle 1)
	Proteomics	Plasma	SciLifeLab	Olink CVDII, CVDIII, Metabolism panels	12,800 (cycle 1)
		Plasma	SciLifeLab	Olink Explore, Proteomics 3072 proteins	5300 (cycle 1 and 300 in cycle 2 SMCC)
	Metagenomics	Stool	Karolinska Institutet	DNBSEQ deep shotgun metagenomic sequencing	7000 (cycle 1 and 2)
	Targeted metabolomics, NMR	Serum	Nightingale Health	About 250 lipoproteins/fatty acids	12,800 cycle 1, 1800 cycle 2 (SMCC)
Ongoing	Targeted metabolomics, LC–MS	Adipose tissue	Chalmers University of Technology	Vitamin D metabolites	12,800 cycle 1, 5000 cycle 2
	Lipidomics, GC–MS/LC–MS	Plasma, Adipose tissue	Örebro University	Lipidomics	12,800 cycle 1, 5000 cycle 2
	Metagenomics	Saliva	Umeå University	Oxford Nanopore Sequencing	40,000 (main cohorts)
Planned	Targeted metabolomics, LC–MS	Plasma	Chalmers University of Technology	200 dietary biomarkers	12,800 cycle 1, 5000 cycle 2
	Untargeted metabolomics, LC–MS	Plasma, Urine, Stool	Chalmers University of Technology		12,800 urine, 7500 stool, 5000 cycle 2 (all types)

LC–MS liquid chromatography-mass spectrometry; *NMR* nuclear magnetic resonance; *GC–MS* gas chromatography-mass spectrometry

[a]40,000 saliva main cohorts, 13,000 blood samples in subcohorts, overlap between blood and saliva samples in 2000 individuals

BeadChip (Illumina Inc, San Diego, CA, USA) for four cancer projects within the OncoArray Consortium [31], including breast [32], ovarian [33], colorectal [34], and prostate [35] cancer. Upon completion of data processing, the SIMPLER genome-wide association study (GWAS) data will include genotype information for approximately 51,000 participants.

Proteomics

Proteomic profiling was conducted in two stages using Olink Proteomics panels (Olink Proteomics, Uppsala, Sweden). First, lithium/heparin plasma samples from all participants in cycle 1 were analyzed using the Olink CVD II, CVD III, and Metabolism panels in 2021. Details of the analytical procedures are provided in Michaëlsson et al. [36]. Second, EDTA plasma from SMCC Uppsala (cycle 1, same participants as above, and 300 from cycle 2) were analyzed using the Olink Explore 3072/384 panel in 2025. These targeted proteomics analyses were conducted at SciLifeLab in Uppsala, Sweden.

Metabolomics

The metabolome profile has been analysed using both untargeted and targeted methods, including mass spectrometry and NMR-based technologies [37].

Untargeted metabolomics

Lithium/heparin plasma samples from cycle 1 were analysed at Chalmers Mass Spectrometry Infrastructure (CMSI, Gothenburg, Sweden) using untargeted reverse-phase ultra-high-performance liquid chromatography-mass spectrometry (UHPLC-MS) on Agilent 6550 Q-TOF mass spectrometers (Agilent Technologies, Inc., Santa Clara, CA, USA) during 2018 and 2019 [38]. A total of 1696 and 3434 clustered metabolite features were recovered in the positive mode in SMCC and COSMC, respectively. In the negative mode, 3187 and 3413 features were detected in SMCC and COSMC, respectively, and work with annotation is ongoing. Additional lithium/heparin plasma samples collected from participants in SMCC and COSMC from examination cycle 2 and cycle 3 will undergo untargeted metabolomics UHPLC-MS at CMSI during 2027–2028.

Targeted metabolomics

In 2023, Nightingale Health (Helsinki, Finland) analysed serum samples from SMCC cycle 1 and cycle 2 for targeted metabolomics using nuclear magnetic resonance (NMR) [39]. In total 250 metabolites or metabolite ratios were measured. During 2026–2028, serum samples from SMCC Uppsala (cycles 1, 2, and 3), and SMCC Västmanland and COSMC (cycles 1 and 2), along with urine, stool, and adipose tissue biopsies, will undergo targeted metabolomics using

UHPLC-MS at CMSI. The analysis aims to detect > 2000 metabolites from > 5000 metabolic features in serum and to assess endogenous and exogenous biomarkers, including dietary biomarkers in plasma and urine.

Lipidomics

Lipidomics analysis [40] using gas chromatography-mass spectrometry (GC–MS)/UHPLC-MS is being performed at Örebro University and data will be available for plasma samples and adipose tissue biopsies in SMCC Uppsala cycle 1 and cycle 2, as well as COSMC and SMCC Västmanland cycle 1 in 2027. Analyses of samples from COSMC cycle 2 and SMCC Uppsala cycle 3 are planned for 2027–2028.

Metagenomics

Metagenomic analysis was performed on stool samples at the Centre for Translational Microbiome Research (CTMR), Karolinska Institutet, Stockholm, Sweden, between 2021 and 2024. Samples to be analysed were aliquoted and refrozen at Uppsala University after adding DNA/RNA Shield buffer and then sent to CTMR for DNA extraction [30]. Sequencing was carried out using 2 × 150 bp paired-end reads on the DNBSEQ G400 or T7 sequencing instrument (MGI, Guangdong, China) at Latvian Biomedical Research and Study Center in Latvia and CTMR, with an average of 51 million reads per sample. Bioinformatic processing of the metagenomic sequencing data was performed using two independent analytical pipelines. First, at CTMR, the StaG-mwc pipeline was applied, including quality control with Fasp, host DNA removal using Kraken2, and taxonomic profiling with MetaPhlAn4, all with default settings [41]. Second, host-filtered sequences from CTMR were further profiled with Clinical Microbiomics Human Microbiome Profiler (CHAMP) at Cmbio (Copenhagen, Denmark) [42, 43].

Following DNA extraction from saliva, a random subset of these samples (n ~ 1100), from participants who also had stool samples, was successfully used in a pilot study conducted at Umeå University, Sweden, to characterize the oral microbiome [44]. Comprehensive analysis of the remaining saliva DNA samples is currently underway, and microbiome data from 40,000 participants is expected to become available in 2027.

Table 3 Number of participants (n) in the Swedish Mammography Cohort (SMC) and Cohort of Swedish Men (COSM) who completed the different questionnaires during the longitudinal follow-up

SMC ♀		COSM ♂	
Questionnaires completed	n	Questionnaires completed	n
1 (1987)	61 433		
2 (1987,1997)	38 984	1 (1997)	45 906
3 (1987, 1997, 2008)	25 250	2 (1997, 2008)	29 503
4 (1987, 1997, 2008, 2009)	21 806	3 (1997, 2008, 2009)	26 006
5 (1987, 1997, 2008, 2009, 2019-Health)	13 207	4 (1997, 2008, 2009, 2019-Health)	15 627
6 (1987, 1997, 2008, 2009, 2019-Health, 2019-Lifestyle)	11 195	5 (1997, 2008, 2009, 2019-Health, 2019-Lifestyle)	13 791

Health health questionnaire; *Lifestyle* lifestyle questionnaire

Results

A total of 190,606 women and men were invited to participate in the SMC and COSM baseline investigations. The number of invited and participating individuals, as well as the response rate at each investigation, is shown in Fig. 2a. More than 4500 questionnaire- and register-based variables were collected per participant during the 30-year study period. In the subcohorts, about 1,400 additional variables per participant were collected, excluding omics data. Table 3 presents the total number of participants who completed each questionnaire. Participation rates were acceptable across all investigations, with initial participation rates of 74% in SMC and 49% in COSM (without a reminder), resulting in more than 100,000 respondents overall. Four questionnaire data collections have been conducted in SMC since 1987–90, and three in COSM since 1997. Data collected in 1997, 2009, and 2019 from both cohorts can be combined and co-analysed, with 84,890 individuals (45,890 men) forming the 1997 combined baseline cohort. Participants in the SMCC and COSMC were mostly randomly selected to provide more in-depth data and biological samples. The third data collection cycle is currently underway for SMCC Uppsala, while the second cycle is underway for COSMC and SMCC Västmanland (Fig. 2b).

Omics data

Omics data in SIMPLER are steadily expanding. Plasma proteomics and metabolomics data are available from ~ 12,800 participants in the subcohorts, genomics data for about 51,000 participants, and metagenomics data is currently available for about 7000 individuals (Table 2). Metabolomics of urine and stool, lipidomics measured in plasma and adipose tissue biopsies, and targeted MS-based plasma metabolomics are not yet available but are expected to be available for use in 2028.

Register data

With more than 30 years of follow-up, corresponding to over 2.5 million person-years, a substantial number of incident disease events and deaths have accumulated in the cohorts. Selected examples of these are presented in Table 4. Follow-up for incident disease cases extends until December 31, 2025. During this period, almost 61 percent of the cohort participants have died, totalling 65,264, including 37,759 women and 27,685 men. For example, 13,052 incident hip fractures and 18,506 stroke cases were recorded. Table 5 summarizes the number of individuals

Table 4 Number of deaths and incident outcomes in SIMPLER consisting of Swedish Mammography Cohort (SMC) and Cohort of Swedish Men (COSM), and their subcohorts, SMC Comprehensive Subcohorts (SMCC) and COSM Comprehensive Subcohort (COSMC)

		Outcome	Main cohorts		Comprehensive subcohorts				Total
						Uppsala	Västmanland		
			SMC ♀	COSM♀	Total	SMCC♀	SMCC♀	COSMC♂	
			61,433	45,906	107,339	5032	2623	4750	12,405
National Cause of Death Register, through Dec 31, 2025		Death	37,579	27,685	65,264	1707	503	1726	3936
National Cancer Register, through Dec 31, 2024		Cancer, cases	22,417	20,345	42,762	1539	431	1429	3399
		Cancer, individuals	18,370	15,995	34,365	1215	363	1150	2728
National Patient Register, through Dec 31, 2025									
ICD 10	ICD 9								
		Total CVD[a]	46,240	35,284	81,524	3469	1608	3556	8633
I20-I25	410–414	IHD	13,302	13,991	27,293	652	259	1116	2027
I21	410	AMI	7435	7702	15,137	310	94	410	814
I60, I61, I63, I64	430, 431, 464, 436	Stroke	10,440	8066	18,506	491	186	514	1191
I110, I50	428	Heart failure	14,702	11,261	25,963	880	312	1017	2209
E10-E14	250	Diabetes	11,120	9,865	20,985	609	319	785	1713
S02, S12, S22, S32, S42, S52, S62, S72, S82, S92	800–829	Any fracture	26,278	12,137	38,415	1798	714	938	3450
S720-S722	820	Hip fracture	9528	3524	13,052	506	187	243	936
NOMESCO codes: CJC, CJD, CJE, CJG, 171, 172, 1733, 1734		Cataract extraction	25,721	14,915	40,636	2664	1180	1733	5577
E	240–279	Endocrine disease	29,179	18,842	48,021	2303	951	1791	5045
N	580–591	Renal disease	34,483	24,232	58,715	2714	1148	2406	6268
M	710–739	Musculo-skeletal disease	37,410	23,724	61,134	3520	1341	2260	7121
D	490–519	Lung disease	30,111	23,717	53,828	2339	1128	2495	5962
S	800–999	Injuries	35,804	21,890	57,694	2647	1075	1926	5648
F	290–319	Psychiatric disease	17,473	10,177	27,650	1193	411	834	2438
G20-G22	294, 332, 736	Parkinson's disease	1411	1427	2838	129	40	157	326
G30-G31	290, 303, 330, 331, 334	Dementia	4362	1956	6318	376	118	249	743
K50-K52	555, 556, 558	IBD	3235	1881	5116	198	117	172	487
K70-K77	571, 572, 573	Liver disease	1360	1094	2454	78	26	77	181
K80-K82	574, 575, 576	Gallbladder disease	6524	3,474	9998	320	86	177	583
K85	577	Acute pancreatitis	1165	941	2106	71	18	57	146
K40-K46	550, 552, 553	Hernia, abdomen	7061	10,594	17,655	490	232	839	1561

Outcomes were collected in the National Cause of Death Register, National Cancer Register, or National Patient Register. Outcomes were collected from baseline in the respective cohorts. Includes main and contributing diagnoses

CVD cardiovascular diseases; *IHD* ischemic heart disease; *AMI* acute myocardial infarction; *IBD* inflammatory bowel disease

[a]Total CVD includes all diagnoses I00–I99

Table 5 Cumulative number of individuals with prescribed drugs, categorized by therapy type (A-S), collected from the Swedish National Prescribed Drug Register in the Swedish Mammography Cohort (SMC) and Cohort of Swedish Men (COSM), and their subcohorts, SMC Comprehensive Cohorts (SMCC) and COSM Comprehensive Cohort (COSMC)

	Main Cohorts			Comprehensive subcohorts			
				Uppsala	Västmanland		
	SMC ♀	COSM ♂	Total	SMCC ♀	SMCC ♀	COSMC ♂	Total
	61,433	45 906	107 339	5 032	2 623	4 750	12 405
A—Alimentary tract and metabolism	42,591	29 455	72 046	4 194	2 184	3 566	9 944
B—Blood and blood-forming organs	42,464	33 633	76 097	3 964	1 974	3 985	9 923
C—Cardiovascular system	47,019	36 401	83 420	4 529	2 356	4 331	11 216
D—Dermatologicals	26,079	18 858	44 937	3 047	1 540	2 507	7 094
G—Genitourinary system and sex hormones	28,124	21 070	49 194	3 575	1 721	3 095	8 391
H—Systemic hormonal preparations[a]	27 095	17 691	44 786	2 776	1 416	2 069	6 261
J—Antiinfectives for systemic use	46 800	35 358	82 158	4 777	2 471	4 401	11 649
L—Antineoplastic and immunomodulating agens	8 071	7 872	15 943	979	467	1 005	2 451
M—Musculo-skeletal system	38 579	27 876	66 455	4 138	2 213	3 726	10 077
N—Nervous system	48 791	35 959	84 750	4 679	2 426	4 217	11 322
P—Antiparasitic products, insecticides and repellents	0	0	0	0	0	0	0
R—Respiratory system	39 976	29 428	69 404	4 312	2 236	3 715	10 263
S—Sensory organs	33 998	20 923	54 921	4 109	1 867	2 811	8 787

Collected from register initiation in 2005 through December 31, 2025, https://sdb.socialstyrelsen.se/if_lak/val_eng.as,

[a]Excluding sex hormones and insulin

with prescribed drugs from the Swedish National Prescribed Drug Register, categorized by therapeutic class. Notably, nearly 80 percent of cohort participants received at least one antibiotic prescription during follow-up (therapy type J) and at least one prescription drug from therapy type N, including sleeping pills and antidepressants.

Discussion

SIMPLER constitutes a unique research resource, providing longitudinal data of exceptional breadth and depth to address major public health challenges and is recognized as a "gold mine" for research. Research based on this infrastructure has generated more than 1000 peer-reviewed, high-quality original publications since the 1990s. In addition, numerous systematic reviews and meta-analyses have utilized data and results from the SIMPLER database, and approximately 80 PhD students have used or are currently using the infrastructure as part of their scientific training. About 100 research projects based on SIMPLER are currently ongoing. Several international consortia have invited SIMPLER to contribute genetic, health, and register data to support analyses of large datasets from diverse populations worldwide [30, 45–53]. More recently, the availability of multi-omics data has further increased interest from national and international users.

Expanded scope

Since the initiation of the cohort studies, the research scope has progressively expanded. As a research infrastructure, SIMPLER provides researchers with unique opportunities to lead or participate in competitive international research across multiple scientific domains. Whereas earlier studies in the cohorts primarily focused on overall mortality, cancer, and cardiovascular diseases, the cohorts now constitute a valuable resource for research on a wide range of additional outcomes. These include infectious diseases, blood diseases, endocrine diseases, mental and neurological diseases, respiratory and gastrointestinal diseases, skin diseases, musculoskeletal disorders, renal diseases, and injuries [54–58]. Over time, an increasing number of outcomes have accrued among participants, enabling epidemiological investigations with high statistical power and without loss to follow-up other than emigration (<0.5%). The SIMPLER database contains a large number of phenotype data repeatedly collected from a large population sample that includes women and men living in central Sweden, thus enabling analyses of updated information exposure and covariates spanning over decades. Both cohorts were representative of the Swedish population with respect to age distribution, body mass index, educational level [2], and smoking status [59] at the 1997 investigation, supporting generalization of the results to the broader Swedish population. It is important to note that baseline data for women in the SMC were collected in 1987–90, whereas for men in COSM the baseline was 1997. Using 1997 as a common baseline in combined analyses therefore means that, for women, this represents follow-up

rather than true baseline data, while for men it is baseline. This difference may introduce bias. However, from 1997 onward, identical questionnaires were used in both cohorts, which enables joint analyses of men and women, a strength of the research infrastructure. Finally, participant representativeness mainly affects studies of disease occurrence but has less impact on analyses of risk factor–disease associations, which are the primary focus of many studies conducted within SIMPLER.

State-of-the-art research

In primary and secondary prevention research, variability in genetic susceptibility to chronic diseases constitutes a key component of the evidence-based precision health. Data from the SIMPLER cohorts can be leveraged for this purpose, and integration with additional omics data from multiple biological samples further strengthens the approach by addressing both clinical and public health concerns. Many chronic late-onset diseases are multifactorial, and their causes are often not fully understood. Molecular pathology diagnostics and disease sub-classification are increasingly applied in clinical translational medicine using genomics, metabolomics, and proteomics. Comprehensive omics characterization of participants in SMCC and COSMC, from genome to exposome, enables refined phenotype definitions and improves understanding of disease aetiology. This approach also enhances insights into disease pathways, thereby supporting advances in prevention, early disease detection, and the categorization and treatment of diseases.

Furthermore, integrated omics data facilitate the identification of new biomarkers and provide deeper insights into disease development, shared aetiologies across diseases, impact of specific lifestyle exposures, and risk prediction. Omics analyses can be used to discover biological mechanisms and pathways. For example, proteomics studies in the SIMPLER cohorts have identified novel associations of circulating proteins with body fat mass [60], incident cardiovascular disease [61], incident hip fracture [36], dietary patterns [62], meat consumption [63], and physical activity [64], thus providing insights into mechanisms and pathways.

Avenues of research

Integrating omics data with information on environmental exposures and health outcomes enables a comprehensive characterisation of molecular complexity. The large number of biomarkers identified across different omics datasets in the cohorts opens multiple opportunities for novel research. Microbiome data, in particular, support hypothesis-driven investigations of the interactions between the environment, microbial communities, and health. For example, stool sample analysis can be used to assess antibiotic resistance by incorporating data on antibiotic use from the Swedish Prescribed Drug Register [65]. When combined with questionnaire-based lifestyle information, these data advance precision health and personalized medicine research. In addition, saliva DNA is available from about 40,000 individuals, allowing not only for human genomic analysis but also for profiling of the oral microbiota. In collaboration with Umeå University, pilot studies have been successfully completed [44], and a larger project is currently underway to characterize the microbiota of the oral cavity in these 40,000 individuals.

The extensive phenotypic data collected within SIMPLER enable validation of GWAS-generated genetic associations, including evaluation of genetic scores in relation to disease development. Gene-environment interaction analyses are essential to establish future individualized lifestyle recommendations. SIMPLER data also support causal inference research [66], including Mendelian randomization studies [67]. Using both GWAS and Mendelian randomization approaches, a study in SIMPLER established a strong causal effect of fibroblast growth factor-21 (FGF21) on improved lipid profile, reduced alcohol consumption, C-reactive protein concentrations, and liver function biomarkers, including fibrosis [68]. Given to the longitudinal design with repeated measurements of exposures, causal mediation analysis and emulation of clinical trials with the initiation of theoretical interventions can be performed, including methodology to emulate the effect of dietary adherence [69]. A causal mediation analysis concluded that a Mediterranean-type diet affects hip fracture risk via pathways other than through type 2 diabetes and body mass index [70].

The apparent lack of protective associations between certain healthy foods and risk of certain cancers, cardiovascular diseases (CVD), and mortality in epidemiological studies may be due to the masking of the association by the concurrent presence of food contaminants, for example, in fish [71] or inadequate adjustment for dietary context and confounders [72]. Estimated dietary exposure to organochlorinated pollutants (e.g., polychlorinated biphenyls; PCBs), a common contaminant in fatty fish, was associated with an increased incidence of various CVDs and CVD-mortality [73]. In contrast, the long-chain polyunsaturated fatty acids found in fish were associated with a lower risk. The organochlorine and CVD associations were corroborated using biomarkers of exposure in a nested case–control study in SMCC [74]. Thus, such studies allow for benefit-risk research by disentangling potential positive and negative effects.

The extensive range of variables available within SIMPLER makes the cohorts well-suited for research on potential

work-related and environmental exposures that may affect long-term health and longevity. Exposures such as noise, air pollution, greenness, chemicals, metals, occupational vibration exposure, and job strain may be included in such studies. When applicable, combining diet data from the cohorts in SIMPLER with occurrence data on environmental contaminants enables estimates of each participant's exposure to these compounds (exposome). It could be assessed in relation to the development of chronic disease. Such studies have been performed, for example on cadmium, pesticide residues, and polychlorinated biphenyls (PCBs). Information on the area of residence facilitated studies on drinking water content [75, 76]. Biomarkers of various contaminants have also been measured in subsamples of the subcohorts and published data have been used by different stakeholders, including the European Food Safety Authority, to derive reference values. Studies have shown that exposure to the toxic metal cadmium was related to, for example, fracture incidence [77, 78]. Another study investigated the impact of air pollution, road traffic noise, and greenness (collected from register data) on the risk of myocardial infarction among women and found that residential greenness was inversely related to risk, while road traffic noise was not [79].

Improvement of diet-disease association studies

Substantial gains in life expectancy can be achieved through improved diets; eating more legumes, whole grains, and nuts, and less red and processed meats, yielded the greatest gains [80]. Data collected in SMC and COSM have long been used in nutritional epidemiology to investigate associations between diet (nutrients, specific foods, and dietary patterns) and the development of chronic diseases, to inform nutrition promotion and prevention strategies [81–87]. Over time, dietary patterns have become increasingly diverse, a trend captured by the expanded FFQs used in the cohorts. However, the habitual diet is a complex set of intercorrelated exposures, and self-reported tools, such as FFQs and dietary records, are subjected to random and systematic errors [88]. These limitations can partly be addressed using biomarkers. Alkylresorcinols (AR) are biomarkers that reflect dietary whole-grain intake. Using data from the SIMPLER cohorts, long-term whole-grain rye but not whole-grain wheat intake was found to be reflected in AR measured in adipose tissue [89]. Biomarkers derived from metabolomics provide an additional complement or alternative to self-reported dietary data, helping to reduce reporting errors and misclassification biases [90, 91]. Metabolomics captures both exogenous dietary exposures and endogenous metabolic responses, offering a comprehensive picture of individual dietary effects and exposure. Biomarkers reflecting specific food components within dietary patterns can also be used to calibrate estimates of dietary intake. In the SIMPLER database, targeted metabolomics-derived food biomarkers measured in plasma and urine will be available in the future.

Future perspectives

As a research infrastructure, SIMPLER provides a unique integration of longitudinal lifestyle data, register information, and multi-omics data from diverse biological samples, which are continuously updated. This comprehensive resource enhances its attractiveness to researchers and strengthens its capacity to address key scientific questions from individual, societal, and public health perspectives. A future objective is to further enrich SIMPLER by incorporating data from additional cohorts, such as WICTORY (Westmannia Cardiovascular Riskfactor study), which includes 34,000 women and men in Västmanland County born between 1940 and 1959. These participants were recruited in health screenings between 1990 and 1999 the year the participants turned 40 or 50 years old [92]. About one-third of these individuals overlap with participants in SMC and COSM. SIMPLER and the Northern Sweden Health and Disease Study (NSHDS) [93] are planned to be integrated into a shared research infrastructure, SIMPLER 2U. NSHDS is a large population-based cohort with biobank samples from approximately 150,000 unique participants, long-term follow-up, and well-documented high-quality samples. For approximately 65,000 individuals, repeated blood samples are available. Another long-term goal is to expand SIMPLER into a multigenerational cohort by inviting the 174,357 children of SIMPLER and WICTORY participants to participate in home-based sample collection, and lifestyle and health questionnaires. Looking ahead, the expanded and enhanced data resources in SIMPLER will support more targeted analytical approaches. Integration of emerging molecular technologies with longitudinal lifestyle data and traditional biomarkers in large-scale cohorts offers new opportunities to improve disease prediction, deepen understanding of disease mechanisms, and strengthen prevention and treatment strategies.

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was written by Eva Warensjö Lemming, Karl Michaëlsson, and Diem Nguyen, and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Declarations

Competing interests The authors have no relevant financial or non-financial interests to disclose.

Ethical approval The study is performed in line with the principles of the Declaration of Helsinki. More than 100 ethical approvals have been obtained over the years, initially from regional ethics committees and, since 2019, from the national Swedish Ethical Review Authority.

Consent to participate For SMC and COSM, participant consent was structured such that completion and return of the questionnaire constituted written informed consent. The completion of the questionnaire was preceded by a two-page information sheet about the study's purposes and data responsibility. For SMCC and COSMC, each participant provided written informed consent on a separate form at each visit.

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